

The Five Safes as a Privacy Context

James Bailie*

bailie@chalmers.se

Chalmers University of Technology and University of
Gothenburg
Gothenburg, Sweden

Ruobin Gong

ruobin.gong@rutgers.edu

Rutgers University
Piscataway, New Jersey, USA

Abstract

The Five Safes is a framework used by national statistical offices (NSO) for assessing and managing the disclosure risk of data sharing. It can be understood as a specialization of a broader concept—contextual integrity—to the situation of statistical dissemination by an NSO. We demonstrate this by mapping the five parameters of contextual integrity onto the five dimensions of the Five Safes. We also discuss how each of these two theories can address weaknesses in the other, thereby strengthening them both.

Keywords

Contextual integrity, statistical disclosure control, official statistics.

1 Introduction

As a supplier of official data, a national statistical office (NSO) is an integral part of a well-functioning democratic state. Its data are essential for informing government policy, business strategy and academic research, thereby advancing society and driving economic growth [13]. Yet an NSO’s ongoing value depends upon maintaining its social license to collect and share data. It is therefore necessary that NSOs balance their social and economic utility with the privacy of their data providers. With the recent increase in resources available to malicious actors and the growth of competing data vendors, this trade-off is increasingly difficult to manage [4].

The Five Safes (FS) is one tool that assists NSOs in navigating data sharing decisions in the presence of potential privacy risks. Originally developed in 2003 to enable researchers’ access to Office of National Statistics (ONS) microdata [8], its use has since expanded to all forms of statistical dissemination [3]. It has been employed by NSOs in the UK, Australia, and New Zealand to guide design decisions and risk assessments for statistical disclosure control (SDC) [3, 11, 20–22]. In the US, the FS have been used by the Coleridge Initiative for sharing data within and across states, government agencies and researchers [9]. Further, the Advisory Committee on Data for Evidence Building recently recommended that the use of the FS in the US federal bureaucracy be expanded [1].

Broadly speaking, the FS is a framework for designing and assessing modes of statistical data sharing in ways that maintain the privacy and confidentiality of data providers. More specifically, it is a theory that decomposes disclosure risk into five dimensions, or ‘safes’—safe *people*, *projects*, *settings*, *data*, and *outputs*—along with a loose set of guidelines for how an NSO can manage risk by appropriate choices in each of these five dimensions. These two parts of the FS are often conflated in existing work, resulting in

considerable confusion surrounding what exactly the FS comprises [10].

In this work, we bring the perspective of contextual integrity (CI) to isolate and explicate the first—and most central—part of the FS: its theory of disclosure risk. We make two main points. Firstly, the FS theory may be viewed as a specialization of CI for statistical dissemination contexts. We support this claim by translating between the five CI parameters (sender, recipient, subject, information type and transmission principles) and the five ‘safes.’ Viewed this way, the FS provides specialized guidance as to how NSOs can satisfy contextual information norms, overcoming some of the difficulties in applying CI to statistical dissemination.

Conversely, by situating the FS within the CI theory, NSOs benefit from the latter’s extensive literature in justifying and understanding the FS. Our second main point, therefore, is that both the FS and CI can benefit from cross-pollination between the two fields, ultimately leading to better—i.e., safer and more useful—data access. Indeed, by stress testing our translation between CI and the FS, our analysis reveals ambiguities in the FS that are addressed by CI, and *visa versa*.

We conclude this article by discussing the FS as a principled yet necessarily incomplete response to the wicked problem of disclosure control, and by outlining future work on its relationship to CI, formal privacy criteria, and the legal frameworks shaping NSO practice.

2 Contextual integrity for statistical dissemination

CI is a theory that provides a structured way of understanding when an information flow preserves privacy: namely, when the broader context of the flow conforms to legitimate social norms [15]. CI posits that five parameters govern such norms: the *sender*, the *recipient*, the *subject(s)*, the *information type*, and the *transmission principle* [17].

Of the five CI parameters, the first two are the simplest to understand. The ‘sender’ of a flow is the actor sending the information and the ‘recipient’ is the actor receiving the information. The third parameter, the ‘subjects,’ are those entities to whom the information pertains: often persons, sometimes businesses, and possibly even other entities, such as families or farms. The fourth parameter, the ‘information type,’ refers to the nature of information transmitted: demographics, medical history, financial records, to name a few possibilities. It is the information’s metadata—the description of how the information was collected and processed, or, in statistical terms, the information’s *generating mechanism*. Finally, the fifth parameter, the ‘transmission principle,’ captures the conditions of the information flow—e.g., the permitted uses, the consent (or lack

*This work was partially undertaken while JB was a PhD student at Harvard University in Cambridge, Massachusetts, USA.

thereof) of the flow, whether the information was purchased, and so on.

The dissemination of statistical information by an NSO operates under a set of distinctive informational norms. In democratic societies, NSOs typically act pursuant to the authority granted by an elected governing body to administer censuses and surveys. The resulting products are disseminated in the public interest to provide a quantitative basis that supports evidence-based research and policy decisions. The legal, regulatory, public policy and social confines within which NSOs conduct statistical dissemination entail a class of *contexts* within the meaning of the CI theory. A common characteristic, or *information norm*, associated with statistical dissemination contexts that we consider in this work is that the information flow be *statistical*, defined as follows.

DEFINITION 1. *An information flow is statistical if the sender intends for it to describe a population of interest, rather than the specific entities from which the information was derived, notwithstanding the fact that those entities may serve as representatives for the population.*

As an illustrative (but incompletely specified) example of a statistical information flow (SIF), suppose an NSO transmits survey responses to a researcher. Even though these responses concern only a sample of households (the *representatives*), the NSO intends that the researcher uses this information to understand the *population* from which the sample was drawn.

Since it qualifies the purposes of an information flow, the term ‘statistical’ is a part of a flow’s transmission principle. In a SIF, the transmitted data must pertain to population-level patterns, like those that can be inferred by statistics or machine learning. This is not to say that individual-level information cannot be shared in a SIF. Indeed, every SIF transmits some individual-level information because information on a population is simply an aggregate of individuals’ information, by virtue of every population being simply a collection of individuals. Moreover, SIFs can transmit unaggregated data pertaining to a single subject (e.g., one individual’s census response, or software diagnostics collected from one user), as well as aggregate data drawn from multiple subjects (e.g., a regression model’s coefficient, or the weights and biases of a neural network). The determining quality here is that the purpose of a census response or a software diagnostic is to learn general information, in this case demographics and common software bugs. That is to say, the qualifier ‘statistical’ is a condition on an information flow’s transmission principle—it is a restriction on how the transmitted information is intended to be used.

The strength of CI lies in its generality and broad applicability. It has been used to explain complex privacy dynamics across a wide variety of settings [15]. However, this same generality makes it difficult to operationalize CI for some types of information flows, especially SIFs [5, 16]. This has resulted in the concept of the transmission principle carrying much of the explanatory and normative burden within the framework. Yet, teasing out the transmission principle from the context of a given information flow is often challenging: rather than being formally codified, it is often implicitly hidden within the social practices, roles, relationships and tacit understandings that structure a context. Consequently, too much of CI’s power hinges on how this parameter is elicited, whereas a more specialized theory—such as the FS—can provide additional

scaffolding for identifying the transmission principle in specific contexts—such as those pertaining to statistical dissemination.

Another limitation of CI is that it assumes that the subjects of any information flow are always clearly delineated from ‘non-subjects.’ Moreover, one must be able to specify the subjects of any given information flow, at least in principle. These assumptions are challenged by many SIFs—particularly modern ones [16]—where the transmitted information is separated from the original data providers by a sequence of complex data processing steps (such as those found in machine learning). The problem is that a SIF often pertains to an ill-defined population, rather than a specific group of entities. As a topical example of this, consider specifying the subjects of the information contained in a large language model. In comparison to CI, the FS theory makes no requirements that the subjects must be well specified (although in SIFs where the subjects are known, they would be included as part of a FS assessment).

3 The Five Safes and the information flows they govern

Much as CI posits that five parameters govern the integrity of an information flow, the central thesis of the FS is that five dimensions of data access—*people*, *projects*, *settings*, *data*, and *outputs*—collectively characterize the disclosure risk of a statistical dissemination. These five dimensions are collectively termed the five ‘safes’ and are best reviewed by explicating the two complementary types of SIFs encountered in statistical dissemination:

data → people, (I)

outputs → general public. (II)

In the flow of Type I, an NSO purveys detailed *data* to a limited set of *people*—e.g., a group of social scientists. Typically, these researchers must first submit an application outlining the specific *project* they are working on and the data they want to analyze in this project. The NSO then determines exactly what data these people can access, and under what *settings* that access is given. After the researchers access and analyze these data, they submit a second application to publish their findings. The NSO decides what *outputs* can be released, which are then disseminated to the general public in a flow of Type II. Alternatively, Type II flows may occur when the data custodian directly shares information derived from their data with the general public, without involving researchers as an intermediary (as in Example 1 below). Analogously, but less commonly, Type I flows may occur without a related flow of Type II, such as when an NSO makes microdata available for download to a trusted set of users (Example 2), who then do not disseminate their findings to the general public.

The FS assesses the disclosure risk of all these kinds of statistical dissemination: those that involve both types of SIFs, as well as those that only involve a single SIF (of either type). Much like CI, the FS’s parameters are related yet independent: a change in one need not affect the others. However, viewing these five dimensions under a joint framework allows a data custodian with a fixed amount of resources to focus on the safety of a subset of these dimensions, while maintaining control of the overall disclosure risk. The framework also allows them to weaken controls in some dimensions, at the expense of stricter controls in other dimensions—e.g., allowing

	Public use data files/ open data	User registered data access	Physical data enclaves	Synthetic data with validation
Type I flow	—	Data are transmitted to registered users	Microdata are accessed within the enclave	Synthetic data are shared with approved users
Type II flow	Open data are disseminated online	Users may publish their findings	Summary outputs are exported from the enclave	Validation results may be published
People	NA*	Medium	High	Medium
Projects	Very low	Low	High	High
Settings	Very low	Low	Very high	Medium
Data	Low	High	Low	Medium
Outputs	Very high	Low	High	High

Table 1: Simplified and heuristic Five Safes assessments for four different modes of statistical dissemination. To illustrate the application of the Five Safes, we rate the NSO's degree of control over each safe across four dissemination modes from very low to very high. Descriptions of these modes are given in Examples 1-4 respectively. *The dissemination of public use data files involves a single SIF of Type II, hence the dimension 'people'—which is a property of Type I SIFs—is not applicable.

access to very detailed microdata to a trusted set of people under a high degree of oversight (see Example 3 below).

Given a statistical dissemination, the 'people' parameter of the FS measures the security of the actors receiving data in the dissemination's Type I flow. The NSO might assess security through the actors' commitment to data confidentiality and research ethics (including compliance with any stated requirements of privacy protection pertaining to the data they access), their qualifications or credentials to handle these data and their institutional backing, if any.

The 'projects' dimension evaluates whether the intended use of the data is appropriate, ethical, and compliant with relevant legislation or regulations. The use of certain sensitive data may be restricted by law to support independent scientific research only [9, Section 12.3]. The data custodian would often also ascertain that their data are used toward the advancement of science, with clear and positive social benefits and in a manner consistent with modern scientific norms, including standards of reproducibility and knowledge sharing. They might also request the project's approval from relevant institutional ethics review boards.

The 'settings' parameter concerns the environment in which the SIFs take place, be they physical or virtual. The key question in this dimension is, how strong are the settings in terms of their ability to restrict impermissible use, while still enabling data access for legitimate recipients? Settings include the various security measures an NSO can employ—e.g., compliance monitoring, automated auditing, firewalls, passwords and two-factor authentication, functional separation, physical or virtual data enclaves (Example 3), validation servers (Example 1) and so on.

The 'data' dimension captures the type, sensitivity and granularity of the information transmitted in the Type I flow of the statistical dissemination. It also concerns the use of any statistical methods—such as data perturbation—to mask the personal information shared in this flow. Salient questions in this dimension include: Do the data represent a sample or a census, and if it is a sample, how was that sample selected? How much identifying information do the data contain, and how likely is it that a researcher would

spontaneously recognize the subject of a data? Were the data generated by linking two or more database, or was it collected via a single instrument? When the statistical dissemination is a single flow of Type II, 'data' describes the type and sensitivity of the microdata that generated the 'outputs' (e.g., the census responses that aggregate to the disseminated population counts).

Finally, the 'output' parameter of the FS is defined analogously to the 'data' parameter, except applied to the information transmitted in the Type II flow (instead of the Type I flow). While the data in a Type I flow might consist of microdata records, the output in Type II flows are frequently aggregate statistics like population counts that have been protected through the addition of a controlled amount of random noise.

In principle, a statistical dissemination could be assessed against each of the five 'safes' on a continuous scale. Yet, as we will discuss in Section 7, these dimensions are difficult to measure quantitatively; in practice an NSO instead has a list of possible choices for each of the five safes, ordered by their strength in reducing disclosure risk. The remainder of this section will provide four common examples of these different choices. These examples are summarized in Table 1.

EXAMPLE 1 (PUBLIC USE DATA FILES/OPEN DATA). *Statistical agencies publish public use data files for access by the general public and researchers alike (a flow of Type II without an accompanying Type I flow). Agencies do not vet people who access these files because, by design, any person or entity without abusive intentions should be able to access these public resources. In contrast, a high level of scrutiny is placed on the outputs—i.e., the information contained in this public data files—to ensure that they are not disclosive. On the other hand, since the data custodian cannot supervise the use of this information once it is made open access, no control is possible regarding the safety of the projects nor the settings in which the projects will be conducted, beyond requiring users to sign a contractual terms of use agreement.*

Public use data files are frequently in the form of tabular data, which are highly aggregated to ensure adequate confidentiality. Public use microdata exist too, but they are often heavily subsampled. The US Census Bureau (USCB) curates the Public Use Microdata Sample (PUMS) based on a small sample (1% and 5%) of responses

from the American Community Survey [24]. The PUMS files are available on the USCB’s website and may be accessed via the file transfer protocol (FTP), a microdata analysis tool, or through an API provided by the bureau.

EXAMPLE 2 (USER REGISTERED DATA ACCESS). *NSOs also make some of their data products available for download by registered users (the Type I flow). These users are vetted by the NSOs, allowing for these products to be slightly more detailed compared to what could be made publicly available. However, apart from requiring registered users to agree to terms of use, NSOs have no direct control on how the users store or share these data. Additionally, the NSOs do not explicitly control the release of the outputs of each users’ analyses (the Type II flow) in this type of dissemination, as the users are free to directly share their findings with the general public.*

Access to the Basic Microdata products from the Australian Bureau of Statistics (ABS) are restricted to registered users. Once registered, they can download these microdata to their own computers. Because this means the data can leave the ABS’s secure IT infrastructure, they highly restrict the type, granularity and detail of data available through this dissemination mode. Some of the registered users may access a Basic Microdata product, but not share the results of their analysis. In this case, there is a SIF of Type I, but not of Type II.

EXAMPLE 3 (DATA ENCLAVES). *Data enclaves are secure access environments through which authorized researchers can query a custodian’s database (the Type I flow). Data enclaves provide a highly secure setting for data access. Both the people and the projects seeking access are heavily vetted: only researchers who demonstrate legitimate scientific purposes of their inquiry and compliance with research ethics are allowed access. The outputs that the researcher is allowed to obtain and bring outside of the data enclave (the Type II flow) is subject to various degrees of scrutiny. As a result, the data accessible through enclaves can be detailed and comprehensive.*

Data enclaves may be physical or virtual. A physical enclave is synonymous with a research data center, such as the Federal Statistical Research Data Center (FSRDC) of the USCB, or the Canadian Research Data Centre Network of Statistics Canada. A virtual data enclave allows authorized researchers to access restricted-use data by logging into a secure, remote server. The DataLab of the ABS is an example of a virtual data enclave.¹ The reader is referred to [3] for further illustrations of dissemination paradigms discussed in Examples 1-3 and an analysis of important safety considerations that pertain to them.

EXAMPLE 4 (SYNTHETIC DATA WITH VALIDATION SERVERS). *The data custodian releases a synthetic dataset that resembles the underlying confidential dataset (the Type I flow). Researchers who hold permission to access the synthetic dataset may use it to compose their desired statistical analysis including its code implementation. Then, they may validate their results (via a “validation server”) with the data custodian, who will run the analysis on the confidential dataset and release the outputs to the researcher (and consequently the general public) (the Type II flow). Before releasing these outputs, the NSO will*

typically apply additional controls—such as adding random noise—to ensure that they are safe.

A prominent case of Example 4 is the Survey of Income and Program Participation (SIPP) Synthetic Beta (SSB) [23]. The SSB is synthesized by the USCB through integrating nine annual SIPP panels between 1984 and 2008, together with the W-2 records from the Social Security Administration and Internal Revenue Service. Researchers whose proposed analysis is deemed appropriate and feasible by the USCB are permitted to access the SSB. Prior to October 2022, access could be obtained via the Synthetic Data Server (SDS) hosted by Cornell University.² Once the researcher composes a statistical analysis program using the SSB, they submit this code to the USCB, which in turn performs the validation on the researcher’s behalf on the Gold Standard File (GSF)—the confidential microdata. The output is subject to a stringent level of disclosure review similar to those applicable to the FSRDCs.

Example 4 is an interesting mode of statistical dissemination, through which we can observe a juxtaposition of safety levels pertaining to the two SIFs. In the Type II flow, the receivers are the general public and the outputs are strictly scrutinized according to a high safety standard, rendering this setting similar to the open data mode discussed in Example 1. In the Type I flow, the transmitted data consists of the SSB, which commands a level of safety higher than simply sharing the GSF due to its synthetic nature. The people, here referring to the researchers who access the SSB, are placed under a moderate level of scrutiny. The setting, the Cornell SDS, is effectively a virtual enclave. These elements render this setting analogous to the data enclave mode discussed in Example 3.

4 The Five Safes as a privacy context for statistical dissemination

The previous section argued that A) a statistical dissemination by an NSO consists of a SIF of Type I followed by a related SIF of Type II, or a single SIF of either type; and that B) the five ‘safes’ are parameters of these information flows. In this section, we embed the FS theory into the broader framework of CI by making the following claim:

CLAIM 1. *The FS is a (faithful albeit inexact) specialization of CI to the dual information flows of statistical dissemination.*

To unpack this claim, we map the meanings of the five CI parameters in statistical dissemination contexts onto the five ‘safes.’ This map is summarized in Table 2. We characterize the FS as a ‘faithful albeit inexact’ specialization because while there some ambiguities in this map, the general character of CI is maintained. Although an attentive reader will no doubt pick up on these ambiguities in what follows, we postpone their discussion to Section 5.4 where we stress test Claim 1.

Of the five CI parameters, the first two are straightforwardly mapped to the FS. The ‘sender’ in a flow of Type I is the NSO, or more generally, the statistical agency or custodian of the data being disseminated. In Type II flows, the ‘sender’ is either this same NSO, or the ‘people’ who received the ‘data’ in the corresponding Type I flow. That there are two options here reflects the fact that there are

¹Due to a lack of full oversight on the data access setting compared to physical data enclaves, statistical agencies debate the safety of virtual data enclaves [see e.g., 19].

²Unfortunately, the Cornell server was shut down on September 30, 2022.

Informational norm parameters	Their meanings in statistical dissemination
sender	NSOs (I) and people or NSOs (II)
recipient	people (I) and general public (II)
subject(s)	is a component of data (I)
information type	is a component of data (I) and outputs (II)
transmission principle	encompasses projects , settings , and more

Table 2: The parameters of contextual informational norms and their meanings in statistical dissemination, with reference to the Five Safes (in bold) and relevant SIFs (Type I or II). We use the term NSOs as shorthand for statistical agencies and data custodians more generally.

two possible scenarios: either the researchers (i.e., ‘people’) share their findings directly with the general public (as in Example 2), or their outputs are released by the NSO (as in Examples 3 and 4). The ‘recipients’ of a statistical dissemination are the ‘people,’ for the Type I flow, and the general public, for the Type II flow.

The latter three parameters of CI require more explication. On face value, the two information flows of a statistical dissemination share the same ‘subjects,’ since they are both based on the same underlying data. Nevertheless, as the ‘data’ shared in a Type I flow is often census or survey questionnaire responses, it is usually easier to understand who its subjects are by associating each response with the entities it measures. In comparison, it can be difficult to determine the ‘subject(s)’ of a Type II flow, as the ‘outputs’ are often the result of a statistical analysis concerning a larger underlying population that is represented by the subjects of the corresponding Type I flow. For these reasons, the concept of ‘subjects’ is often not appropriate for Type II flows [16], but usually falls within the ‘data’ dimension for Type I flows. That said, the FS does not require the subjects of a Type I flow to be specified, and there are cases where such a requirement would be nonsensical—e.g., when releasing synthetic data (Example 4).

Because the two flows of a statistical dissemination transmit different kinds of information, they each have their own ‘information type.’ The ‘information type’ of a Type I flow is a component of ‘data’ since the nature of the information transmitted concerns both the structure and the substance of the ‘data.’ On the other hand, the ‘information type’ of a Type II flow is a component of ‘outputs’ because the custodian’s decision about what to release to the general public includes a determination on its information type.

For a statistical dissemination, the transmission principle encompasses two dimensions of the FS: ‘projects’ and ‘settings.’ As described in Section 3, to assess whether a project is safe is to ask whether the dissemination is used for an appropriate purpose—be it to support scientific research, serve the public interest, or enable evidence-based policy making—and to ensure compliance with legal and regulatory mandates. The intended use of the dissemination delineates the scope of the project, and in this capacity, the purpose partially determines the transmission principle. Similarly, ‘safe settings’ are concerned with the environment of the dissemination—e.g., whether it is conducted in a data enclave. Clearly, these settings are also conditions on how the information flows from NSO to the ‘people’ or general public and as such, would be classified under CI as part of the transmission principle.

Until now, we have followed the standard presentation of the FS as a risk assessment tool. In contrast, CI is a normative theory—it is concerned with explaining and describing privacy in terms of social norms. While practitioners of the FS would typically ask questions of the form, “Are the values of the five ‘safes’ sufficient to ensure that the risk of a privacy violation is acceptably low?”, the standard question in CI is, “Are the values of the CI parameters appropriate in the sense that they conform to ethically legitimate contextual informational norms?”

On first glance, these differing perspectives pose a challenge to our claim that the FS is a specialization of CI (Claim 1). Yet, in the case of statistical dissemination, these perspectives are two sides of the same coin. The ethical responsibility of an NSO is to mitigate disclosure risk while simultaneously supplying access to its data in the promotion of the public good. As such, an NSO’s disclosure risk assessment is precisely a question of whether it is meeting its normative duties. This reasoning is the basis for the following claim:

CLAIM 2. *The five ‘safes’ can be viewed as the parameters of the contextual informational norms that define the appropriateness of a statistical dissemination.*

Because it reorients the FS as a normative theory, Claim 2 underpins our view of the FS as a specialization of CI (Claim 1). It is supported by the idea that the FS is not simply a risk assessment tool, but also a framework for trading off the utility of an NSO’s data assets with the consequent disclosure risks. For example, it is well recognized in the FS literature that a project that is of great public interest can offset riskier choices for the other four ‘safes.’ This kind of trading off of competing interests is not valid from the perspective of risk assessment (as public interest clearly does not reduce risk); rather, it is an example of the calculus used in CI for evaluating the legitimacy of privacy norms [15].

5 What can contextual integrity do for the Five Safes?

5.1 GROUNDING THE FIVE SAFES IN NORMATIVE THEORY

The FS is primarily described as a risk assessment tool. As we have argued (see Claim 2), however, it may equally be viewed as a theory for describing the privacy norms of a statistical dissemination. Recasting the framework in this way allows the FS to draw on insights from the CI literature, which offers tools for understanding and evaluating the appropriateness of an information flow (see Section 5.2 below). Furthermore, CI brings a normative dimension

that can legitimize an NSO's choices for the five 'safes' by anchoring them in shared expectations about acceptable dissemination practices, rather than in abstract notions of risk alone.

Grounding the FS in a normative theory helps elucidate the best practices with its application. The FS framework has been criticized because it is unclear how risks from across its five dimensions should be aggregated into a holistic measure. Indeed, [7] argues that a single vulnerability in any one of the five 'safes' can render the dissemination unsafe. Regardless of the validity of this argument, framing the FS as a normative theory offers a way to sidestep this difficulty. Rather than attempting to compute an overall measure of risk, the framework can be seen as articulating conditions under which information flows are socially, legally, or normatively appropriate. In this light, it could be more productive to treat the FS not as a model of risk, but as a normative framework of appropriate statistical dissemination, based on the existing theory developed by CI.

5.2 ELICITING APPROPRIATE CHOICES FOR THE FIVE 'SAFES'

The integration of CI's empirical toolkit with the FS offers a principled means of aligning statistical disclosure practices with societal expectations. CI has been used extensively to elicit and measure privacy norms and expectations, both through quantitative approaches such as vignette-based surveys [14] and through qualitative methods such as interviews and focus groups [12]. These methods enable researchers to evaluate the appropriateness of specific information flows.

By considering the FS as a normative theory, NSOs can apply these techniques to elicit socially grounded choices for each of the five 'safes.' Rather than relying solely on expert judgment or procedural tradition, decisions about data access, researcher vetting, project legitimacy, data treatment, and dissemination channels could be informed by evidence of publicly endorsed norms of appropriateness.

5.3 FRAMING STATISTICAL DISSEMINATION AS INFORMATION FLOWS

CI's concept of an information flow is useful for describing the FS because it makes explicit the dual SIFs of statistical dissemination. Expressing statistical dissemination in this way makes it clear that the FS concerns two separate, yet related, SIFs. Building the framework of the FS through these dual SIFs facilitates risk assessments in much the same way that the decomposition of risk into five 'safes' does.

Furthermore, framing the FS in terms of SIFs separates its theoretical core from its practical application. In practice, the FS is operationalized through guidelines—heuristics and principles for assessing risk in each of the five 'safes.' Most criticisms of the FS target these guidelines, without recognizing that it is the theory of the FS that is the more fundamental and, we contend, more useful part of the framework [7, 25]. By clearly separating theory from guidelines, we hope to clarify critiques of the FS and to refocus attention away from debates over the validity of particular guidelines, thereby enabling improvements to both its theoretical framework and its practical guidance.

5.4 UNCOVERING AMBIGUITIES IN THE FIVE SAFES

By analyzing the FS from the perspective of CI, we identify three ambiguities in its theory—that is, three aspects of an NSO's statistical dissemination that are not considered explicitly by the FS. The identification of these ambiguities clarifies the scope of the FS and in so doing helps strengthen its theory. It also illustrates the fundamental limitations in mapping the FS onto CI (i.e., it qualifies Claim 1).

Firstly, a FS assessment does not explicitly specify the sender of a Type II flow. In some situations, this sender will be the NSO—e.g., when the NSO releases public use data files (Example 1). In other situations—e.g., when trusted users can download and analyze microdata (Example 2)—the sender will be equal to the 'people' parameter of the FS assessment. And finally, the sender of a Type II flow could alternatively be both the NSO and the 'people' acting together. This can occur when the 'people' is working inside a data enclave (Example 3), as they typically must negotiate with the NSO about what outputs will be released (e.g., the NSO may give the researcher a 'budget,' which they can 'spend' on releasing more outputs of lower accuracy, or fewer outputs of higher accuracy). Although not captured by the FS, clearly, from the perspective of CI, it matters which out of the three options the sender of a Type II flow is.

Secondly, CI recognizes that there are many different ways to 'release outputs to the general public'—and indeed, many different ways to interpret what the 'general public' means. There are evident differences between publishing an output statistic on the front page of the New York Times as compared to burying it at the end of a hundred page policy document, even if the general public can access the policy document. How outputs are disseminated to the public matters. This nuance is missed by the FS because it does not distinguish between the different ways of publishing statistics. Moreover, it is not always the case that 'people' will disseminate their findings to the general public. For example, a registered user (Example 2) could share their outputs only with a business interest group. This changes the 'receiver' of the Type II flow from the general public to this interest group. Therefore, should the FS theory make explicit two sets of 'receivers'—one for the Type I flow and one for the Type II flow?

Thirdly, it is important to recognize that the FS is a *self*-assessment. The 'sender' of the Type I flow is not explicitly stated, as it is taken as given that the NSO undertaking the assessment is the entity disseminating the data. Consequently, the 'sender' is not treated as a parameter within the FS, while it is in CI. There are also parts of the transmission principle that are not treated by the FS because they are assumed to be fixed. For one, a flow is always taken to be statistical (in the sense of Definition 1). Moreover, the operating environment of the NSO—e.g., its governmental or legislative legitimacy and its social license, or lack thereof—is not explicitly addressed in the FS.

However, the idea that the FS should always be viewed as a self-assessment is complicated by the fact that NSOs frequently acquire administrative data from other government agencies (e.g., as part of the SIPP SSB outlined after Example 4). In such cases, the NSO could be understood as an intermediary rather than the sender. In fact, the transfer of administrative records to the NSO should be

viewed as a distinct information flow—a flow which is relevant for assessing the disclosure risk of the statistical dissemination, but which is not captured within the FS framework.

More generally, the theory of the FS could be enriched by leveraging the extensive discussion in CI on the different kinds of transmission principles. This may lead to additional ‘safes’ that more explicitly describe the transmission principle of a statistical dissemination (as proposed in [18]).

6 What can contextual integrity learn from the Five Safes?

6.1 CONNECTING THE DUAL INFORMATION FLOWS OF STATISTICAL DISSEMINATION

As we have described, a statistical dissemination is often comprised of two sequentially dependent flows: a Type I flow, in which data are transmitted to accredited researchers under controlled access arrangements, and a Type II flow, in which derived findings are disseminated to the public. These two flows form a causally and normatively linked system: the legitimacy of the public release depends on the propriety of the underlying access, and conversely, the justification for researcher access is often premised on the social value of eventual publication.

CI offers limited traction for analyzing such multi-stage or coupled information flows. It treats each flow as analytically discrete, even when flows are operationally and ethically entangled. The FS framework, however, is explicitly designed to manage the interdependence of a dissemination’s dual flows by modeling them as a coordinated process in which safeguards and social value are distributed across both flows.

Relatedly, another benefit of the FS lies in its treatment of the information ‘subject(s).’ As discussed in Section 2, CI presupposes the existence of discernible subjects—an assumption often violated by Type II flows, where the connection to individual data providers is oblique or diffuse (see Section 4). The FS offers a pragmatic solution by linking the two flows explicitly: the subjects of a Type I flow are indirectly implicated in the corresponding Type II flow, thereby resolving the difficulty often encountered in specifying the ‘subjects’ of a SIF of Type II.

6.2 EXPLICATING ASPECTS OF THE TRANSMISSION PRINCIPLE IN STATISTICAL DISSEMINATION

As discussed in Section 2, CI’s transmission principle bears a substantial analytical burden—to the point that some scholars have proposed augmentations and extensions to CI [5, 6] to explicate aspects of an information flow that might otherwise be understood through the transmission principle. This makes CI less tractable in specific scenarios, like statistical dissemination, where the relevant flows are structured by well-defined institutional practices and procedural safeguards. In comparison, the FS framework offers a domain-specific instantiation of CI that makes the abstract elements of the transmission principle more explicit and actionable.

In particular, the ‘projects’ parameter foregrounds (among other things) how the dissemination is to be used—a critical determinant of the ethical legitimacy of an information flow [15], which is left implicit in CI’s abstract formulation of the transmission principle. Secondly, the ‘settings’ parameter explicates the IT security controls

put in place by the NSO to protect the dissemination. These two parameters detail aspects of the transmission principle, enabling more fine-grained reasoning about the appropriateness—or the disclosure risk—of a statistical dissemination. However, as noted in Section 5.4, ‘settings’ and ‘projects’ together do not exhaust the full scope of the transmission principle. They provide a structured decomposition relevant to the dissemination context, but they are not a substitute for the broader viewpoint provided by the concept of the transmission principle. This discussion illustrates our larger point that both frameworks—CI and the FS—have much to gain through the cross-fertilization of their separate bodies of literature, a subject that future work could further develop beyond the limited analysis given here.

7 Discussion

7.1 THE WICKED PROBLEM OF DISCLOSURE CONTROL

Disclosure control in statistical dissemination poses a “wicked problem.” Two defining characteristics of a wicked problem are: 1) the relevant stakeholders have conflicting interests; and 2) the problem is so complex that there may not be consensus as to what the problem actually is, let alone a solution to it [2]. Facing the problem from outside of the CI framework, NSOs frequently frame statistical disclosure control as a “privacy-utility tradeoff.” This framing bears witness to both characteristics of a wicked problem. On the one hand, data custodians such as NSOs have a dual mandate to procure high quality data for its users while protecting the privacy of data subjects. The demand from data users (the *recipients* of a SIF) for finer and more accurate information comes into direct conflict with the expectation of data *subjects* for confidentiality and anonymity. (It is important to note that this framing treats privacy as secrecy, or obscurity of information, in direct contradiction to CI theory. For this reason, it is better described as an “*obscurity*-utility tradeoff.”) On the other hand, the fact that the privacy-utility tradeoff has remained as a loosely defined catchphrase demonstrates how successfully it has resisted attempts to be operationalized into a set of workable guidance, let alone one that commands wide agreement.

The FS framework makes a valuable contribution by articulating, and hence clarifying, an otherwise intractable problem. It reduces the “wickedness” of disclosure control decisions by modularizing a statistical dissemination into five manageable dimensions, along which an NSO may assess their risk tolerance and utility preference, subject to the pragmatic constraints they face. Moreover, reframing an NSO’s disclosure control problem in terms of CI theory—as this article takes initial steps toward—moves away from the predominant “privacy-utility tradeoff” formulation. While such a formulation may inject clarity and concreteness into an otherwise ill-posed problem, it is invariably reductive and its framing of privacy is problematic. Taking the perspective of the FS and CI, in contrast, establishes the problem in a way that accounts for its entire context.

It is one thing to articulate a problem, and a whole other to devise a solution, though. A data custodian looking to the FS and the FS only for an answer to their disclosure control question may be disappointed. Its five components do not come equipped with yardsticks that provide quantitative measures of safety for a given dissemination regime. As skeptics of the FS rightly point out [see

e.g., 7], it only provides qualitative narratives that may be inherently unfit as quantitative standards.

All is not lost in our view. The FS has demonstrated its value in instructing NSOs' decision-making by evaluating the risk of existing statistical dissemination paradigms (such as those reviewed in Section 3) in each of the five dimensions. The framework is best viewed as a set of principles to reason with the potential factors that affect the disclosure risks of particular statistical dissemination choices. We may easily concede that these efforts in no way provide absolute measures of privacy for all conceivable dissemination choices, nor watertight guarantees of the absence of counterexamples. Such are the distinct advantages of technical criteria of privacy (e.g., differential privacy) by virtue of their mathematical rigor. We believe, however, that while narrow constructions of the disclosure control problem—such as technical privacy criteria—afford the apparent comfort of a concrete and precise answer, they sidestep the heart of the challenge: fostering a consensus about what the disclosure control problem really is. Indeed, as we will explicate in future work, criteria like differential privacy pertains only to certain aspects of the 'output' and 'data' dimensions. That is to say, the precision and rigor of formal privacy notions invariably render them as insufficient answers to the broad, multidimensional problem of statistical dissemination. To this end, we reiterate our viewpoint that the FS, as a reparametrization of CI, provides a distinct conceptualization of the privacy-utility tradeoff for statistical dissemination, and hence a viable alternative angle for analysis.

7.2 FUTURE RESEARCH

Before closing, we briefly discuss two important questions that remain unanswered in this work. First, our reparametrization of CI by the FS is faithful but inexact, in the sense that each framework captures some aspects of the disclosure control problem missing in the other. In future work, we aim to supplant the meanings that are lost in this translation, in particular by expanding the discussion in Section 6 to detail aspects of 'data' and 'outputs' (from the FS) that go beyond the 'subject(s)' and 'information type' (from CI).

Second, we look to further explore the interaction between the FS and the legal frameworks governing the operations of the NSOs. For starters, we ask: what configurations of the FS best correspond to the current legal framework for a specific NSO? And how can the FS be used to update legal frameworks in the future?

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